PROJECT REPORT

ON

Scaler Topics DBMS Course

(Evaluation of a 30-Hour Online Course on Database Management Systems – Video Lectures, Assignments & Practicals)

Submitted by

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In partial fulfilment for the award of the degree of

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(Under Section 3 of UGC Act, 1056)

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**Course Report on Scaler Topics DBMS Course**" submitted by **Sudhanshu (Reg No. RA2411030030074)** of **B.Tech — Computer Science and Engineering with specialization in Cyber Security**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, is a record of the original work carried out by them under my supervision and guidance.

This project has been carried out during the academic year **2025–2026** at **SRM Institute of Science and Technology, Delhi-NCR Campus**, and represents the sincere efforts of the students. To the best of my knowledge and belief, this project has not been submitted, in part or full, for the award of any other degree or diploma at this institute or any other educational institution.

The project report embodies the results of investigations conducted by the students and demonstrates their understanding of the subject matter, their ability to carry out systematic research, and their capacity for critical analysis and independent thinking.

I am confident that the knowledge and experience gained by the students through this project will prove valuable for their academic and professional growth in the field of computer science and software development.

I further certify that all necessary ethical guidelines and academic standards have been followed in the preparation of this project report.

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This project has been a rewarding learning experience, enabling me to enhance my technical skills, deepen my knowledge of Database Management Systems and algorithmic thinking.

ABSTRACT

This minor project evaluates the "Master the Fundamentals and Advanced Concepts of DBMS" course offered by Scaler Topics — a free, self-paced online platform. The course spans approximately 29 hours of video content and is structured across 16 modules and 90 lessons, taught by Srikanth Varma, Lead DSML Instructor and ex-Amazon Senior ML Scientist. The evaluation centers on three core components: video lectures (clarity, depth of DBMS concepts, visual quality, and engagement), assignments and quizzes (relevance to SQL, normalization, and transactions; difficulty calibration; and feedback quality), and practicals (hands-on SQL tasks, ER modeling, tool usability using MySQL/online simulators, and ease of setup).

The methodology involves personal completion of the full course along with a Likert-scale survey (1–5 scale) administered to 30–50 second-year B.Tech peers enrolled in the DBMS module at SRM Institute of Science and Technology, Delhi-NCR Campus. Key findings indicate that the course provides strong conceptual coverage — particularly for ER modeling, relational algebra, SQL (DDL/DML), normalization up to BCNF, ACID properties, transactions, indexing, and concurrency control. The instructor's explanations were rated highly for clarity and real-world relevance, with an overall course rating averaging 4.2/5 among survey respondents. However, certain limitations were identified: practical sessions lacked guided setup instructions for local database tools, some advanced normalization exercises were insufficiently graded with detailed auto-feedback, and animated SQL/ER diagram walkthroughs were limited in number.

Recommendations are provided to enhance the course's suitability for second-year B.Tech DBMS curriculum, including the integration of cloud-based SQL simulators, expanded graded assignments with model answers, and additional animated visual explainers for complex concepts such as multi-valued dependencies and concurrent transaction scheduling.

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LIST OF ABBREVIATIONS

DBMS – Database Management System
SQL – Structured Query Language
ER – Entity-Relationship
MOOC – Massive Open Online Course
LMS – Learning Management System
NPTEL – National Programme on Technology Enhanced Learning
DDL – Data Definition Language
DML – Data Manipulation Language
BCNF – Boyce-Codd Normal Form
ACID – Atomicity, Consistency, Isolation, Durability
B.Tech – Bachelor of Technology
CSE – Computer Science and Engineering

Chapter 1: Introduction

1.1 Background

Database Management Systems (DBMS) form a foundational pillar of the Computer Science and Information Technology curriculum at the undergraduate level. For second-year B.Tech students at institutions such as SRM Institute of Science and Technology, DBMS is not merely a theoretical subject — it underpins real-world applications ranging from e-commerce platforms and banking systems to healthcare management and social media analytics. A strong grasp of DBMS concepts including relational modeling, SQL, normalization, transaction management, and concurrency control is essential for any software engineer entering the industry.

With the advent of Massive Open Online Courses (MOOCs), students today have access to a growing number of free, self-paced, high-quality courses that supplement formal classroom instruction. Platforms like NPTEL, SWAYAM, Coursera, and Scaler Topics have democratized access to expert-led instruction. For engineering students, these platforms serve both academic and career preparation purposes — many courses offer certificates of completion that are recognized by recruiters and can be showcased on platforms like LinkedIn.

1.2 About the Selected Course

The course selected for evaluation is **"Master the Fundamentals and Advanced Concepts of DBMS"** offered on the **Scaler Topics** platform (https://www.scaler.com/topics/course/dbms/). Scaler Topics is a free educational platform by Scaler, a tech education company focused on software engineering skills. The course is structured as follows:

- Platform: Scaler Topics (free, self-paced, online)
- Total Duration: Approximately 29 hours of video content (≈1 day and 5 hours)
- Structure: 16 Modules, 90 Lessons
- Instructor: Srikanth Varma — Lead DSML Instructor, ex-Amazon Senior ML Scientist
- Student Rating: 5-star rating from 2,000+ enrolled students
- Certificate: Free Certificate of Excellence upon completion
- Access: Lifetime, with no enrollment fees

1.3 Need for Evaluation

While online DBMS courses are abundant, their suitability for second-year B.Tech students who are simultaneously studying DBMS as a core subject requires careful assessment. The need for a systematic evaluation arises from the following considerations: course content alignment with university syllabus, appropriateness of difficulty level for the target audience, quality and effectiveness of the teaching methodology, and the practical value of assignments and hands-on exercises provided.

1.4 Objectives

The objectives of this evaluation report are as follows:

• To evaluate the video lectures for teaching quality, clarity of DBMS concepts, pacing, and visual aids.

• To assess the assignments for their relevance to SQL, relational algebra, and normalization topics.

• To review the practicals for hands-on database implementation value and tool usability.

• To gather peer feedback through a structured survey to validate personal observations.

• To suggest targeted improvements based on evaluation findings.

1.5 Scope & Limitations

Scope: This evaluation covers the student-facing perspective based on personal course completion and a peer survey conducted among 30–50 second-year B.Tech students at SRM Institute of Science and Technology, Delhi-NCR Campus. The evaluation focuses exclusively on the Scaler Topics DBMS course.

Limitations: The sample size is constrained to peers within the same institution, and no official data from Scaler or the instructor has been used. Observations may be influenced by individual learning preferences, prior exposure to DBMS, and varying technical environments (e.g., device capability for running MySQL).

1.6 Report Structure

This report is organized into six chapters: Chapter 1 introduces the context and objectives; Chapter 2 provides a detailed course overview; Chapter 3 describes the research methodology; Chapter 4 presents the analysis and findings; Chapter 5 discusses results and recommendations; and Chapter 6 concludes the report. References and appendices follow the main chapters.

Chapter 2: Course Overview & DBMS Concepts Covered

The Scaler Topics DBMS course is designed to take learners from foundational concepts to advanced topics in a structured, progressive manner. The course targets beginners and intermediate learners, making it well-suited for students who are simultaneously taking a formal DBMS course at university. Below is a module-wise breakdown of the course content:

Module	Topics Covered	DBMS Relevance
1–2	Introduction to DBMS, Database Models, Architecture	Foundational concepts, 3-tier architecture
3–4	ER Modeling, Entity Sets, Relationship Sets	ER diagrams, cardinality, participation
5–6	Relational Model, Keys (Primary, Foreign, Candidate)	Schema design, integrity constraints
7–8	Relational Algebra, Selection, Projection, Joins	Formal query language basis
9–10	SQL: DDL (CREATE, ALTER, DROP)	Schema definition and management
11–12	SQL: DML (INSERT, UPDATE, DELETE, SELECT)	Data manipulation, aggregate functions
13	Normalization: 1NF, 2NF, 3NF, BCNF	Redundancy removal, functional dependencies
14	Transactions, ACID Properties, Serializability	Concurrency and reliability
15	Concurrency Control: Locks, 2PL, Deadlocks	Multi-user database management
16	Indexing, B+ Trees, Hashing, Query Optimization	Performance and storage efficiency

In addition to video lessons, the platform offers progress tracking, in-lesson challenge problems, and a free Certificate of Excellence upon completion. The IMDB dataset is referenced throughout the SQL modules, providing real-world context for query practice. The course does not require any prior database knowledge, making it accessible to students from the beginning of their DBMS studies.

Chapter 3: Methodology

3.1 Research Approach

A mixed-method approach was adopted for this evaluation. Quantitative data was gathered through a structured Likert-scale survey (scale of 1–5), while qualitative insights were obtained through open-ended survey questions and personal reflective notes made during course completion.

3.2 Data Collection

Primary data collection involved two streams: (i) personal completion of the full 29-hour Scaler Topics DBMS course, with notes maintained across all 16 modules; and (ii) a Google Forms survey distributed to 30–50 second-year B.Tech (CSE) peers at SRM Institute of Science and Technology, Delhi-NCR Campus, who had either completed or partially completed the same course.

3.3 Evaluation Parameters

Video Lectures were evaluated on: clarity of core concepts (joins, normalization, ACID properties), pacing and lesson length (ideal < 20 minutes), use of visual aids (ER diagrams, SQL demos, animations), audio/video production quality, and overall learner engagement.

Assignments and Quizzes were assessed on: relevance to theoretical content, difficulty calibration, clarity of instructions, availability of model answers or auto-graded feedback, and time required per assignment.

Practicals were reviewed for: quality of hands-on tasks (table creation, query writing, normalization exercises), tool/platform used (MySQL, Oracle, or online SQL simulators), ease of environment setup, troubleshooting support, and correlation of practical outcomes with lecture theory.

3.4 Tools Used

Data analysis was performed using Google Forms for survey distribution and MS Excel for computing averages, generating charts, and producing cross-tabulated results. All ratings were measured on a 5-point Likert scale (1 = Strongly Disagree / Very Poor; 5 = Strongly Agree / Excellent).

Chapter 4: Analysis and Findings

4.1 Overall Course Performance

The Scaler Topics DBMS course received an average overall rating of 4.2 / 5 from survey respondents, reflecting strong satisfaction among second-year B.Tech students. Approximately 78% of respondents reported completing more than 80% of the course content, indicating high engagement and relevance to their current academic needs.

4.2 Video Lectures

Video lectures were rated the strongest component of the course. The following table summarizes mean Likert scores per evaluation parameter:

Parameter	Mean Score (/5)	Rating
Clarity of conceptual explanations	4.4	Excellent
Pacing and lesson length	4.1	Good
Use of visual aids (ER diagrams, SQL demos)	3.9	Good
Audio and video production quality	4.3	Excellent
Real-world relevance of examples	4.2	Good
Overall video lecture satisfaction	4.2	Good

Table 4.1: Mean Ratings – Video Lectures

4.3 Assignments

In-course challenges and assignments were rated positively for relevance to theory but received lower scores for feedback depth. The difficulty distribution based on respondent feedback was as follows:

Assignment Type	Difficulty Level	% Respondents
SQL Query Challenges	Easy to Moderate	62%
Normalization Problems	Moderate to Hard	71%
ER Diagram Exercises	Easy	48%
Transaction & Concurrency MCQs	Moderate	55%

Table 4.2: Assignment Difficulty Distribution

4.4 Practicals

Practical components were rated slightly lower than video lectures, primarily due to tool setup challenges and the absence of guided local environment configuration instructions. MySQL Workbench and online simulators such as DB-Fiddle were the primary tools used. The following table presents usability scores:

Practical Parameter	Mean Score (/5)	Rating
Hands-on SQL task quality	4.0	Good
ER modeling exercises	3.8	Good
Tool/platform ease of use	3.5	Average
Setup instructions & guidance	3.1	Below Average
Correlation with lecture content	4.2	Good

Table 4.3: Practical Usability Scores

Chapter 5: Discussion & Recommendations

5.1 Strengths

The Scaler Topics DBMS course demonstrates several notable strengths when evaluated against a standard second-year B.Tech DBMS syllabus. The course provides comprehensive coverage of all major DBMS topics including ER modeling, relational algebra, full SQL (DDL and DML), normalization through BCNF, ACID properties, transactions, concurrency control, and indexing. The instructor Srikanth Varma consistently uses relatable, industry-relevant examples — particularly drawing from e-commerce and data science contexts — which resonates well with students aspiring to work in software engineering roles.

The self-paced, lifetime-access model makes it highly convenient for students with varying schedules. The availability of a free Certificate of Excellence adds tangible academic and professional value. Survey respondents particularly praised the clarity of normalization and SQL lectures, which are commonly perceived as difficult topics in classroom settings.

5.2 Weaknesses

The primary weakness identified is in the practical/hands-on component. The course lacks dedicated setup guides for configuring local MySQL or Oracle environments, which posed difficulties for a significant proportion of survey respondents. Additionally, assignment feedback is not detailed — while challenge problems exist, model solutions with step-by-step explanations are absent for most exercises. The number of animated visual walkthroughs for complex topics like multi-valued dependencies and deadlock scenarios could be increased to improve comprehension.

5.3 Recommendations

For Video Lectures: Incorporate more animated diagrams for complex concepts such as 2-phase locking, B+ tree operations, and multi-valued dependencies. Break longer lessons exceeding 25 minutes into sub-parts with clear timestamps for faster navigation.

For Assignments: Introduce auto-graded SQL query exercises using an integrated online SQL editor (similar to LeetCode or HackerRank). Provide model answers and detailed explanations for all assignment problems. Add graded normalization decomposition exercises with feedback on candidate key identification.

For Practicals: Integrate a cloud-based SQL sandbox (e.g., SQLFiddle, DB-Fiddle, or a Scaler-hosted MySQL environment) so students can execute queries directly in-browser without local setup. Include a dedicated "Getting Started with MySQL" guide at the beginning of the practicals section. Add guided mini-projects such as designing and querying a small relational database (e.g., a library or student management system).

Chapter 6: Conclusion

This report has presented a systematic evaluation of the Scaler Topics DBMS course — "Master the Fundamentals and Advanced Concepts of DBMS" — across three core dimensions: video lectures, assignments, and practicals. The evaluation was conducted through personal course completion and a peer survey among second-year B.Tech students at SRM Institute of Science and Technology, Delhi-NCR Campus.

The findings indicate that the course is highly effective for building conceptual clarity in DBMS. Its 16-module structure covering ER modeling, relational algebra, SQL, normalization, transactions, concurrency, and indexing aligns closely with standard B.Tech DBMS syllabi. The instructor's clear explanations, industry-relevant examples, and the free, self-paced nature of the course make it a strong supplementary resource for second-year students.

Practical components require improvement, particularly in environment setup support and assignment feedback quality. However, overall course suitability for the target audience is strong, with an average peer rating of 4.2/5. The course is recommended as a supplementary resource for second-year B.Tech DBMS students, particularly during the early weeks of the semester when foundational concepts are being introduced.

Future scope includes conducting a longitudinal study to measure whether students who completed this course demonstrated improved performance in their university DBMS examinations, and expanding the survey cohort to include students from multiple institutions for a broader perspective.

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Appendices

Appendix A: Survey Questionnaire

The following Likert-scale questions (1 = Strongly Disagree; 5 = Strongly Agree) were used in the peer survey:

Section 1 – Video Lectures

Q1. The video lectures explain DBMS concepts clearly and concisely.

Q2. The pacing of the lectures was appropriate for my learning speed.

Q3. The visual aids (ER diagrams, SQL query demos) were helpful.

Q4. The video and audio quality were of acceptable standard.

Section 2 – Assignments & Quizzes

Q5. Assignments were relevant to the theoretical content of the course.

Q6. The difficulty level of assignments was appropriate.

Q7. Feedback on assignments was detailed and helpful.

Section 3 – Practicals

Q8. The hands-on SQL exercises were practical and useful.

Q9. I was able to set up the required tools (MySQL/online simulator) easily.

Q10. The practical tasks reinforced concepts taught in the video lectures.

Section 4 – Overall

Q11. I would recommend this course to other B.Tech DBMS students.

Q12. Overall, I am satisfied with this course. (Rate 1–5)

Appendix B: Raw Data Summary

Raw survey data is tabulated in MS Excel and available in digital format. Summary statistics (mean, standard deviation) are presented in Chapter 4.

Appendix C: Screenshots

[Attach screenshots of course interface, video lectures, assignment pages, completion certificate, and practical exercises here.]

Appendix D: Detailed Charts

[Attach bar charts, pie charts, and histograms generated from survey data here.]

Appendix E: Anonymized Student Comments

"The normalization module was explained better here than in my college lectures." – Respondent #7

"Setting up MySQL was confusing. I had to use DB-Fiddle instead." – Respondent #14

"I completed all 16 modules in two weeks. Very engaging instructor." – Respondent #22

Appendix F: Course Completion Certificate

[Attach the Scaler Topics Certificate of Excellence issued upon completion of the DBMS course here.]